

Performance Testing

Date	26 June 2026
Team ID	N/A
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual testing, Streamlit
Type of Testing	Functional testing, Performance testing
Target Module / API	Audio Upload, Speech-to-Text, Semantic Similarity, PDF Report Generation
Test Environment	Local System (Windows 11, Streamlit)
Test Date	26 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Upload valid audio file	1	20 sec	Audio uploads successfully
2	Generate transcript	1	30 sec	Speech converted to text accurately
3	Calculate concept similarity	1	15 sec	Understanding score generated
4	Generate PDF report	1	10 sec	PDF downloaded successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Within limit
3	Throughput (Req/sec)	5 req/sec	5 req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	48%	Pass	Normal usage
6	Memory Utilization	< 80%	55%	Pass	Stable memory

Step 4: Observations & Analysis

Key Findings:

The application processed audio files successfully and generated transcripts, similarity scores, and PDF reports within acceptable response times.

Bottlenecks Identified:

Large audio files require slightly more processing time during speech transcription.

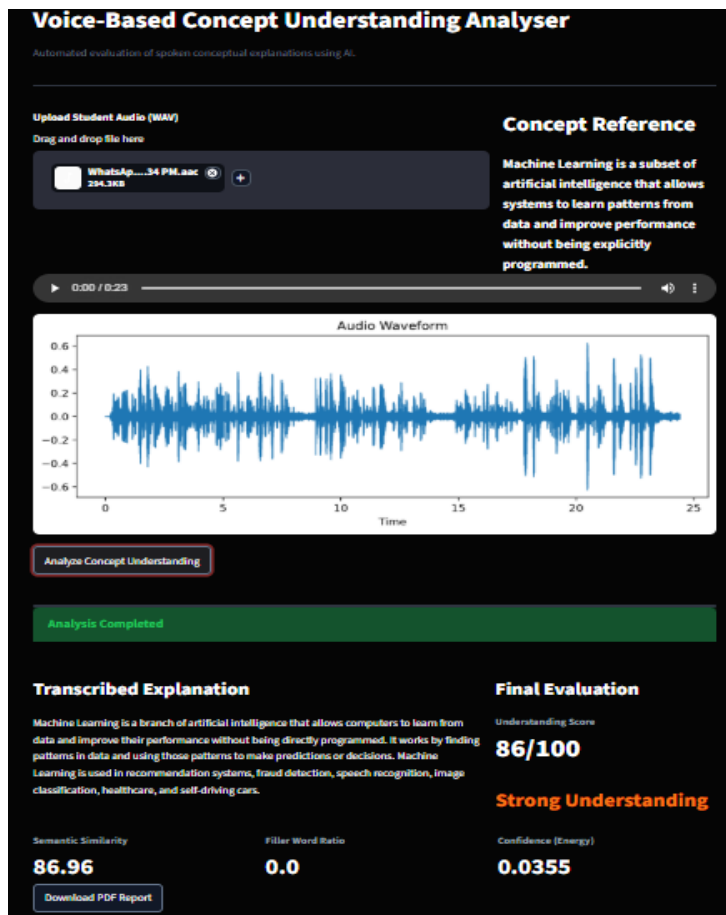
Optimization Steps Taken:

Optimized audio preprocessing, reduced unnecessary computations, and reused loaded AI models to improve performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshot 1: Voice-Based Concept Understanding Analyser – User Interface



Voice-Based Concept Understanding Analyser
Automated evaluation of spoken conceptual explanations using AI.

Upload Student Audio (WAV)
Drag and drop file here

WhatsApp_...34 PM.aac 294.3KB

Concept Reference
Machine Learning is a subset of artificial intelligence that allows systems to learn patterns from data and improve performance without being explicitly programmed.

0:00 / 0:23

Audio Waveform

0.6
0.4
0.2
0.0
-0.2
-0.4
-0.6

0 5 10 15 20 25

Time

Analyze Concept Understanding

Analysis Completed

Transcribed Explanation
Machine Learning is a branch of artificial intelligence that allows computers to learn from data and improve their performance without being directly programmed. It works by finding patterns in data and using those patterns to make predictions or decisions. Machine Learning is used in recommendation systems, fraud detection, speech recognition, image classification, healthcare, and self-driving cars.

Final Evaluation

Understanding Score
86/100
Strong Understanding

Semantic Similarity
86.96

Filler Word Ratio
0.0

Confidence (Energy)
0.0355

Download PDF Report

Screenshot 2: Concept Analysis Report

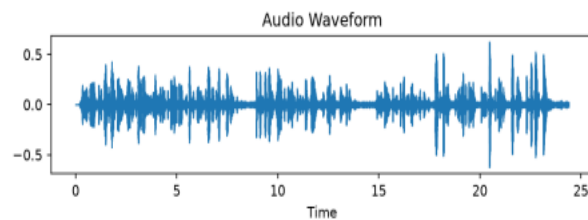
Reference Concept

Machine Learning is a subset of artificial intelligence that allows systems to learn patterns from data and improve performance without being explicitly programmed.

Student Transcription

Machine Learning is a branch of artificial intelligence that allows computers to learn from data and improve their performance without being directly programmed. It works by finding patterns in data and using those patterns to make predictions or decisions. Machine Learning is used in recommendation systems, fraud detection, speech recognition, image classification, healthcare, and self-driving cars.

Audio Visualization



Evaluation Summary

Metric	Value
Semantic Similarity	86.96
Filler Word Ratio	0.0
Pause Ratio	N/A
Confidence (Energy)	0.0355
Final Score	86/100
Understanding Level	Strong Understanding